

Module 01: Systems — Variational Calculus and the Free Energy Functional

Course: Active Inference 401 | **Unit:** Advanced Theory | **Audience:** Graduate students / researchers

Learning Objectives

1. Derive the **free energy functional** from first principles using variational calculus.
2. Analyze the mathematical properties of **variational free energy** — its decomposition into energy and entropy, and its relationship to KL divergence and model evidence.
3. Evaluate the conditions under which **free energy minimization** is equivalent to exact Bayesian inference.

Key Concepts

1. Variational Calculus Foundations

Variational calculus deals with finding functions that optimize functionals — functionals being mappings from function spaces to real numbers. Active Inference uses this framework because the quantity being optimized (free energy) is a functional of a probability distribution $q(s)$, which is itself a function.

The fundamental problem: Given a functional $F[q]$ that maps a probability distribution $q(s)$ to a real number, find the distribution $q^*(s)$ that minimizes $F[q]$.

Euler-Lagrange equation: For a functional $F[q] = \int L(q, q', s) ds$, the minimizer satisfies:

$$\partial L / \partial q - d/ds (\partial L / \partial q') = 0$$

In Active Inference, the Lagrangian L involves the log joint probability (energy) and the entropy of q , and the Euler-Lagrange solution yields the optimal approximate posterior.

2. Deriving the Free Energy Functional

Starting from Bayes' theorem and the concept of model evidence:

Step 1 — Model evidence: For a generative model $p(o, s | m)$ specifying how observations o and hidden states s are jointly distributed under model m :

$$\ln p(o | m) = \ln \int p(o, s | m) ds$$

This integral is generally intractable.

Step 2 — Introduce approximate posterior: Introduce a variational distribution $q(s)$ and use Jensen's inequality:

$$\ln p(o | m) = \ln \int p(o, s | m) \cdot [q(s) / q(s)] ds \geq \int q(s) \ln [p(o, s | m) / q(s)] ds$$

Step 3 — Define free energy: The variational free energy is:

$$F[q] = - \int q(s) \ln [p(o, s | m) / q(s)] ds = E_q[-\ln p(o, s)] + E_q[\ln q(s)]$$

This decomposes as: $F = \text{Energy} - \text{Entropy}$

Step 4 — Relate to model evidence:

$$\ln p(o | m) = -F[q] + D_{\text{KL}}[q(s) \| p(s | o, m)]$$

Since KL divergence ≥ 0 , we have: $F[q] \geq -\ln p(o | m)$

Free energy is an upper bound on surprise (negative log evidence). Minimizing F with respect to q tightens this bound, making q approach the true posterior.

3. Decompositions of Free Energy

Variational free energy admits several illuminating decompositions:

Decomposition 1 — Energy minus Entropy:

$$F = E_q[-\ln p(o, s)] - H[q]$$

The first term is the expected "energy" (negative log joint probability under q). The second term is the entropy of q . Minimizing F balances accuracy (fitting the data) with complexity (keeping q simple/entropic).

Decomposition 2 — Complexity minus Accuracy:

$$F = D_{\text{KL}}[q(s) \parallel p(s)] - E_q[\ln p(o \mid s)]$$

The first term is the complexity cost — how far the approximate posterior $q(s)$ diverges from the prior $p(s)$. The second term is the accuracy — how well q predicts the observations. This decomposition shows that free energy minimization implements automatic Occam's Razor — it penalizes models that deviate too far from their priors.

Decomposition 3 — Surprise plus KL divergence:

$$F = -\ln p(o) + D_{\text{KL}}[q(s) \parallel p(s \mid o)]$$

Free energy equals surprise (neg-log-evidence) plus the approximation error (KL from q to the true posterior). When $q = p(s \mid o)$ exactly, $F = -\ln p(o)$ exactly.

4. Mean-Field and Factorized Approximations

When the exact posterior is intractable, structured approximations constrain q :

Mean-field approximation: Assume q factorizes across groups of hidden states:

$$q(s_1, s_2, \dots, s_n) = \prod_i q_i(s_i)$$

Each factor is updated independently: $q_i^*(s_i) \propto \exp(E_{\{q_{-i}\}}[\ln p(o, s)])$

This iterative scheme (variational message passing) converges to a local minimum of F .

Bethe approximation: Relaxes mean-field by allowing pairwise interactions, corresponding to belief propagation on factor graphs. This is the mathematical foundation for the message-passing algorithms used in Active Inference implementations.

Laplace approximation: Assume q is Gaussian, centered on the MAP estimate. This is computationally efficient and is the basis of Dynamic Causal Modelling (DCM) in neuroimaging.

5. Variational Free Energy on Factor Graphs

When the generative model is represented as a factor graph, variational free energy decomposes into local contributions:

Each factor node f_a contributes a local free energy: $F_a = -E_{\{q(s_a)\}}[\ln f_a(s_a, o_a)] +$ terms involving q

Message passing on the factor graph (variational message passing, belief propagation) implements distributed minimization of total free energy $F = \sum_a F_a$, with messages carrying sufficient statistics between factor nodes.

This provides the bridge between the abstract variational calculus and the concrete implementations of Active Inference.

Summary

The free energy functional is derived from the intractability of Bayesian model evidence, using variational methods to create a tractable upper bound. Free energy decomposes into energy minus entropy (or equivalently, complexity minus accuracy, or surprise plus approximation error). Structured approximations (mean-field, Bethe, Laplace) enable tractable optimization. On factor graphs, free energy decomposes into local contributions enabling distributed message passing.

Further Reading

- Blei, D. M. et al. (2017). Variational inference: A review for statisticians. *JASA*, 112(518), 859-877.
- Beal, M. J. (2003). *Variational Algorithms for Approximate Bayesian Inference*. PhD thesis, UCL.

- Dauwels, J. (2007). On variational message passing on factor graphs. *IEEE ISIT*, 2546-2550.
- Parr, T. et al. (2022). *Active Inference*. MIT Press, Chapters 2-4.